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manchodu@Srini:~/LSLAB$ ./manchi
Simple Shell
Type a command (type 'exit' to quit)

shell> hello
You entered: hello

shell> hellx
You entered: hellx

shell> hellp
You entered: hellp

shell>

shell> exit
Exiting shell...
manchodu@Srini:~/LSLAB$ hell
Command 'hell' not found, but can be installed with:
sudo snap install hell
manchodu@Srini:~/LSLAB$ |
```

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manchodu@Srini:~/LSLAB$ cat shell.c
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define MAX_INPUT 100

/* Enable character-by-character keyboard input */
void enableRawMode(struct termios *original)
{
    struct termios raw;

    tcgetattr(STDIN_FILENO, original);

    raw = *original;

    /* Disable Enter buffering and automatic echo */
    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSANOW, &raw);
}

/* Restore normal keyboard settings */
void disableRawMode(struct termios *original)
{
    tcsetattr(STDIN_FILENO, TCSANOW, original);
}

int main()
{
    char input[MAX_INPUT];
    char ch;
    int index;
    struct termios original;

    enableRawMode(&original);

    /* Main interactive loop */
    while (1)
    {
        index = 0;

        printf("myshell> ");
        fflush(stdout);

        /* Capture keyboard input */
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    read(STDIN_FILENO, &ch, 1);

    /* ENTER KEY */
    if (ch == '\n' || ch == '\r')
    {
        input[index] = '\0';
        printf("\n");
        break;
    }

    /* BACKSPACE KEY */
    else if (ch == 127 || ch == 8)
    {
        if (index > 0)
        {
            /* Remove character from buffer */
            index--;

            /* Remove character from screen */
            printf("\b \b");
            fflush(stdout);
        }
    }

    /* NORMAL CHARACTER */
    else if (index < MAX_INPUT - 1)
    {
        input[index] = ch;
        index++;

        printf("%c", ch);
        fflush(stdout);
    }
}

/* EXIT CONDITION */
if (strcmp(input, "exit") == 0)
{
    printf("Exiting shell...\n");
    break;
}

/* EMPTY INPUT */
if (index == 0)
{
    continue;
}

```

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/* BACKSPACE KEY */
else if (ch == 127 || ch == 8)
{
    if (index > 0)
    {
        /* Remove character from buffer */
        index--;

        /* Remove character from screen */
        printf("\b \b");
        fflush(stdout);
    }
}

/* NORMAL CHARACTER */
else if (index < MAX_INPUT - 1)
{
    input[index] = ch;
    index++;

    printf("%c", ch);
    fflush(stdout);
}
}

/* EXIT CONDITION */
if (strcmp(input, "exit") == 0)
{
    printf("Exiting shell...\n");
    break;
}

/* EMPTY INPUT */
if (index == 0)
{
    continue;
}

/* Display corrected/final input */
printf("You entered: %s\n", input);
}

disableRawMode(&original);

return 0;
}

```